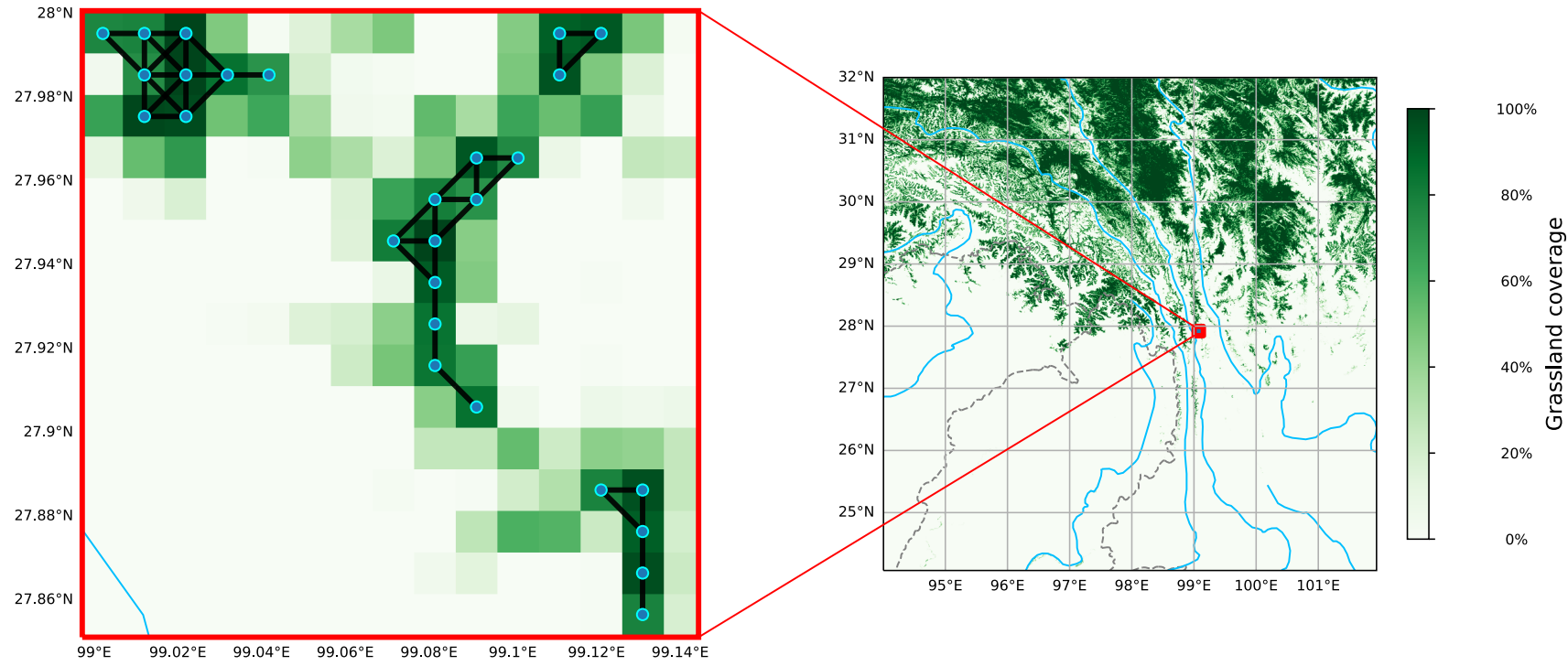


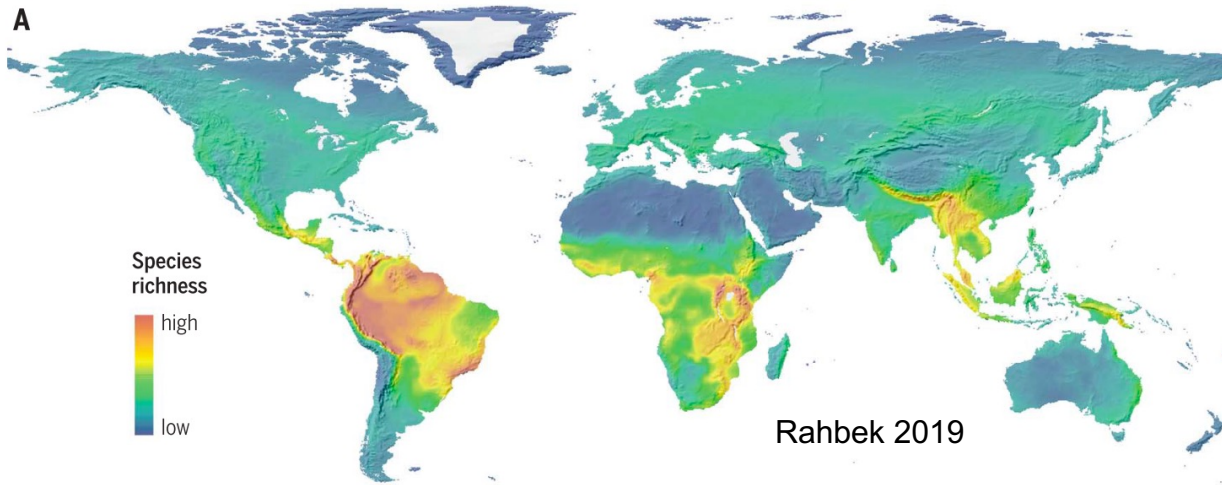


Using graph-based metrics to assess the effect of landscape topography on diversification

Victor Boussange, Loïc Pellissier, and Yaquan Chang

ECBC Amsterdam 2021

Large-scale geographical patterns of species diversity



Mountains represent **25 % of land area**, but **85% of the world's species** of amphibians, birds and mammals, many entirely restricted to mountains (Rahbek 2019)

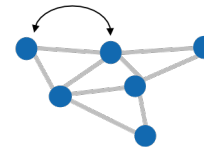


Topological constraints **Habitat heterogeneity**

Underlying mechanisms?



First principles modelling approach



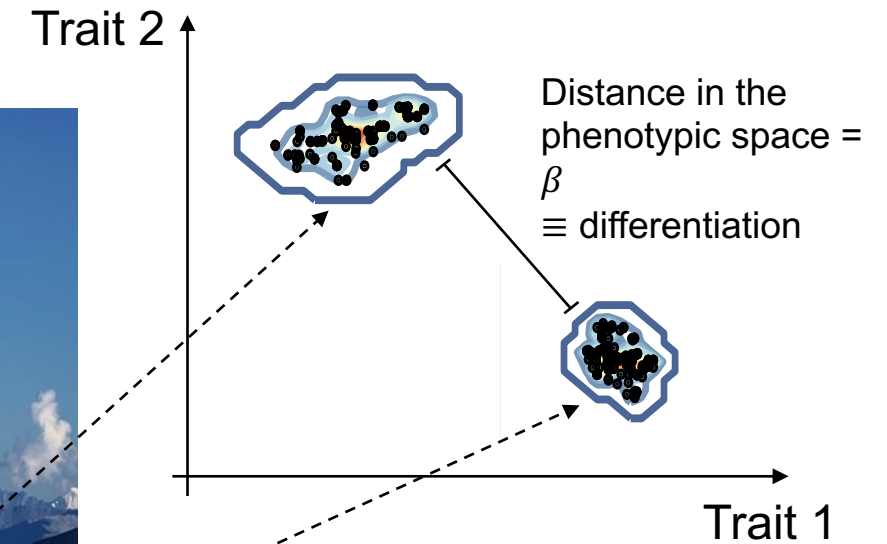
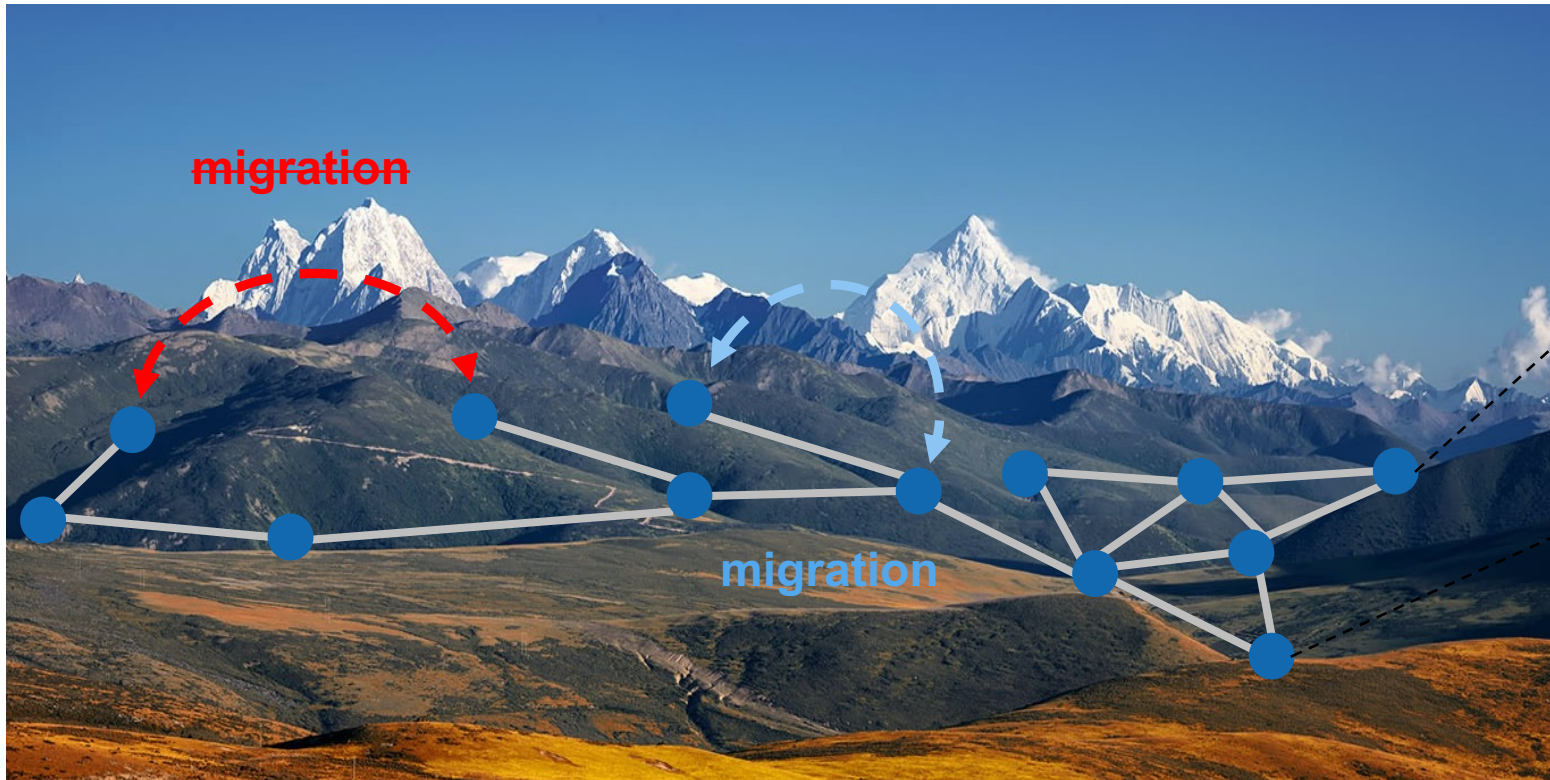
Graph representation
of the landscape

$$m\partial_t v = \sum_i F_i$$

Eco-evolutionary individual
based model

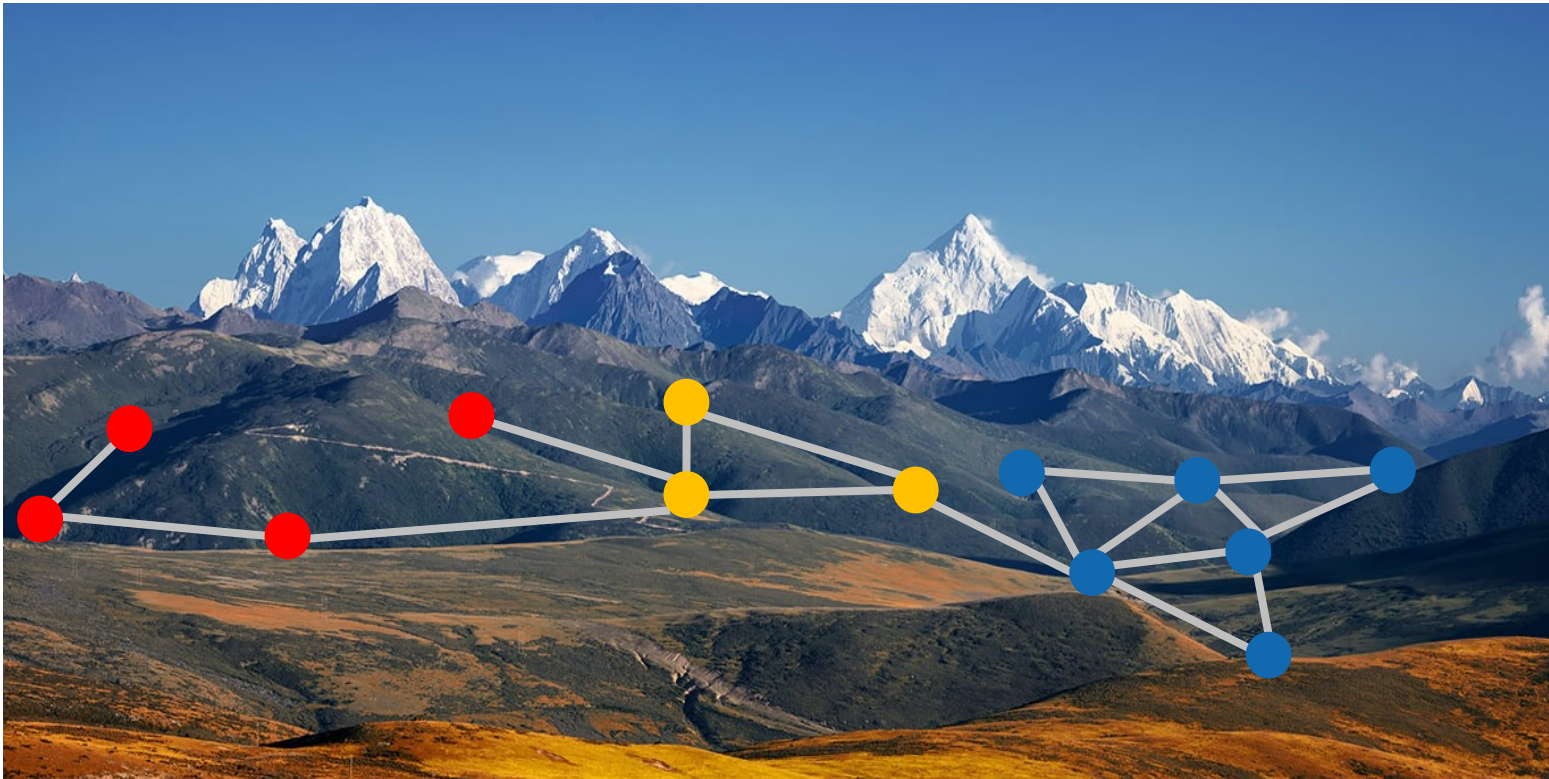
Capturing complex connectivity patterns with graphs

Graphs, to capture dispersal patterns
→ Geometric constraints



Capturing habitat heterogeneity with graphs

Graphs, to capture
environmental heterogeneity



- Habitat 1
- Habitat 2
- Habitat 3

Graph-based eco-evolutionary model

Numerical simulations

&

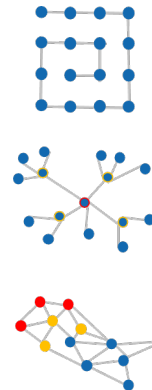
Analytical results



$$\begin{aligned}
 L\phi(\nu_t^{(i)}) = & \int_{\mathcal{X}} \left\{ b_i(\mathbf{x})(1-\mu)(1-m)(\phi(\nu_t^{(i)} + \delta_{\mathbf{x}}) - \phi(\nu_t^{(i)})) \right\} \nu_t^{(i)}(d\mathbf{x}) \\
 & + \int_{\mathcal{X}} \left\{ \mu(1-m) \int_{\mathcal{X}} b_i(y)(\phi(\nu_t^{(i)} + \delta_z) - \phi(\nu_t^{(i)})) \mathcal{M}(\mathbf{x}, y) dy \right\} \nu_t^{(i)}(d\mathbf{x}) \\
 & + \iint_{\mathcal{X}} \left\{ \frac{1}{K} (\phi(\nu_t^{(i)} - \delta_{\mathbf{x}}) - \phi(\nu_t^{(i)})) \nu_t^{(i)}(dy) \nu_t^{(i)}(d\mathbf{x}) \right\} \\
 & + \sum_{j \neq i} \frac{a_{i,j}}{d_j} \int_{\mathcal{X}} \mu m \left\{ \int_{\mathcal{X}} b_j(y)(\phi(\nu^{(j)} + \delta_{\mathbf{x}}) - \phi(\nu^{(j)})) \mathcal{M}(\mathbf{x}, y) dy \right\} \nu_t^{(j)}(d\mathbf{x}) \\
 & + \sum_{j \neq i} \frac{a_{i,j}}{d_j} \int_{\mathcal{X}} \left\{ b_j(\mathbf{x})(1-\mu)m(\phi(\nu^{(j)} + \delta_{\mathbf{x}}) - \phi(\nu^{(j)})) \right\} \nu_t^{(j)}(d\mathbf{x}).
 \end{aligned}$$



- Characteristic length
- Heterogeneity in degree
- Environmental assortativity

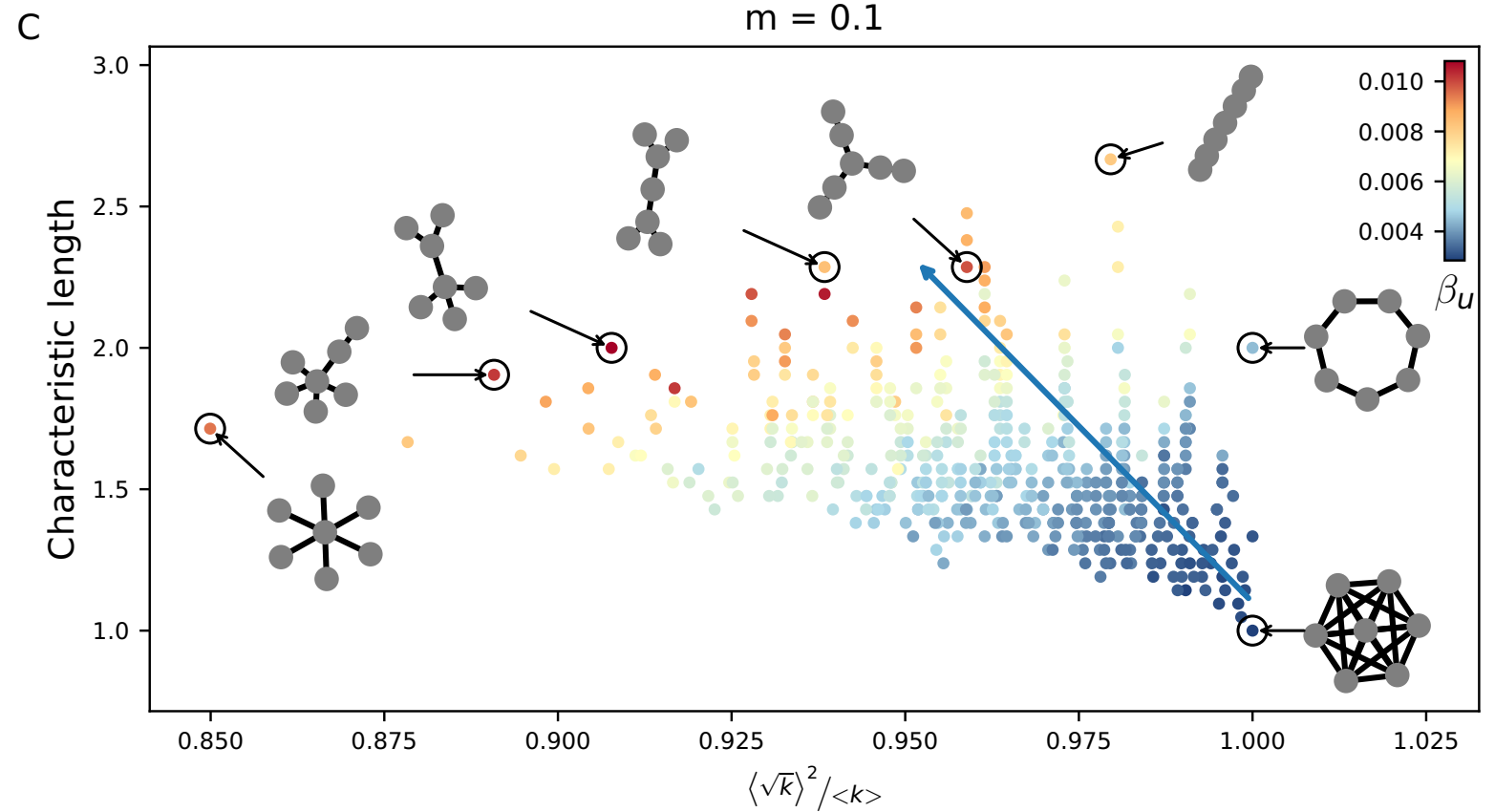
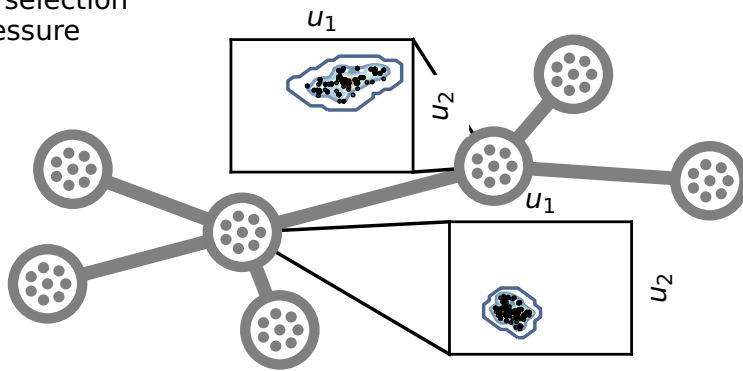


Mechanisms

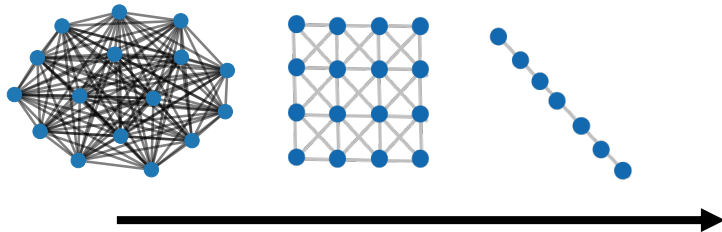


Experiment #1 – Effect of topology on differentiation

Setting (1):
no selection
pressure



Characteristic length and heterogeneity in degree explain 88% of diversity variation



Characteristic length
~ landscape dimensionality

average shortest path between all pairs of nodes in the graph

Nodes with relatively high degree



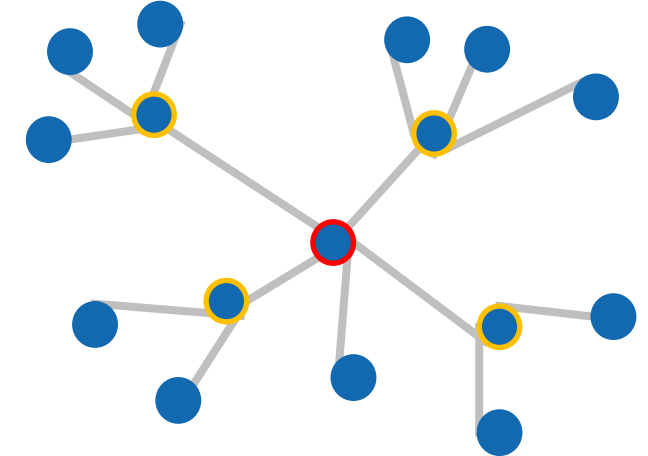
High influx of migrants



Increased competition



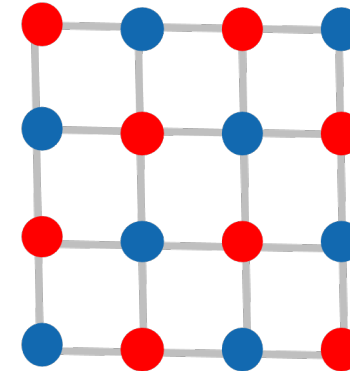
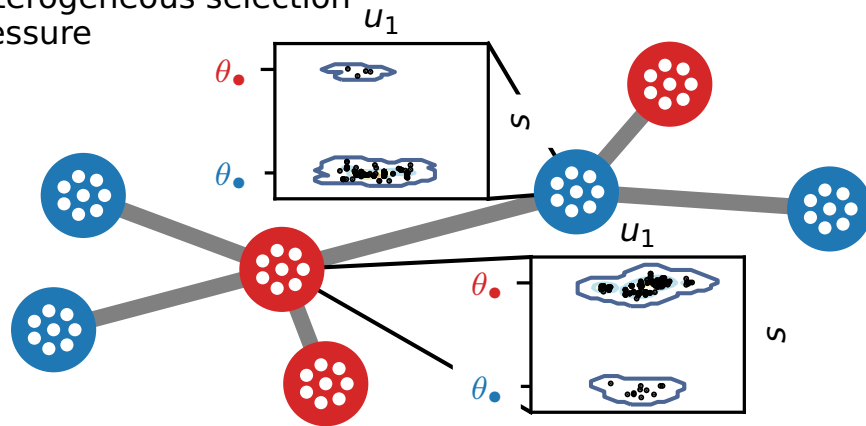
Higher death rate



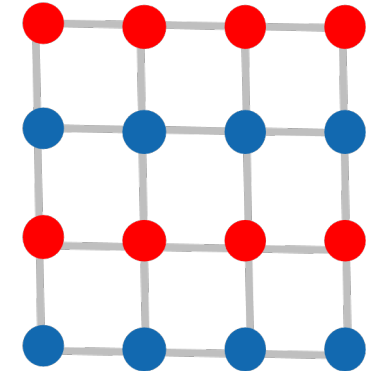
Bottlenecks
Barriers to dispersal

Experiment #2 – Effect of topology on adaptive differentiation

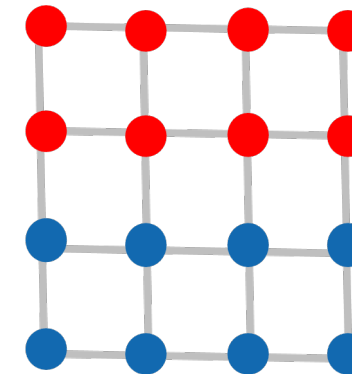
Setting (2):
heterogeneous selection
pressure



$$r_\theta \approx -1$$



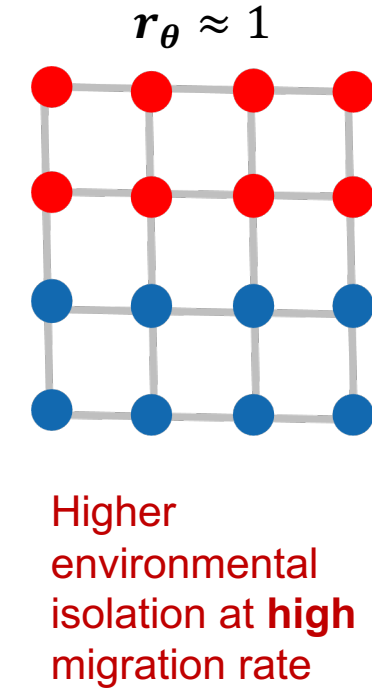
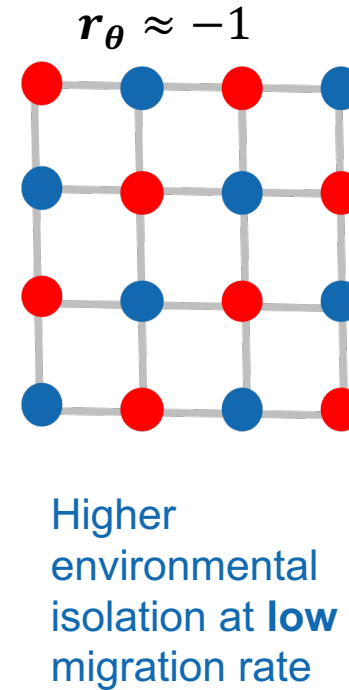
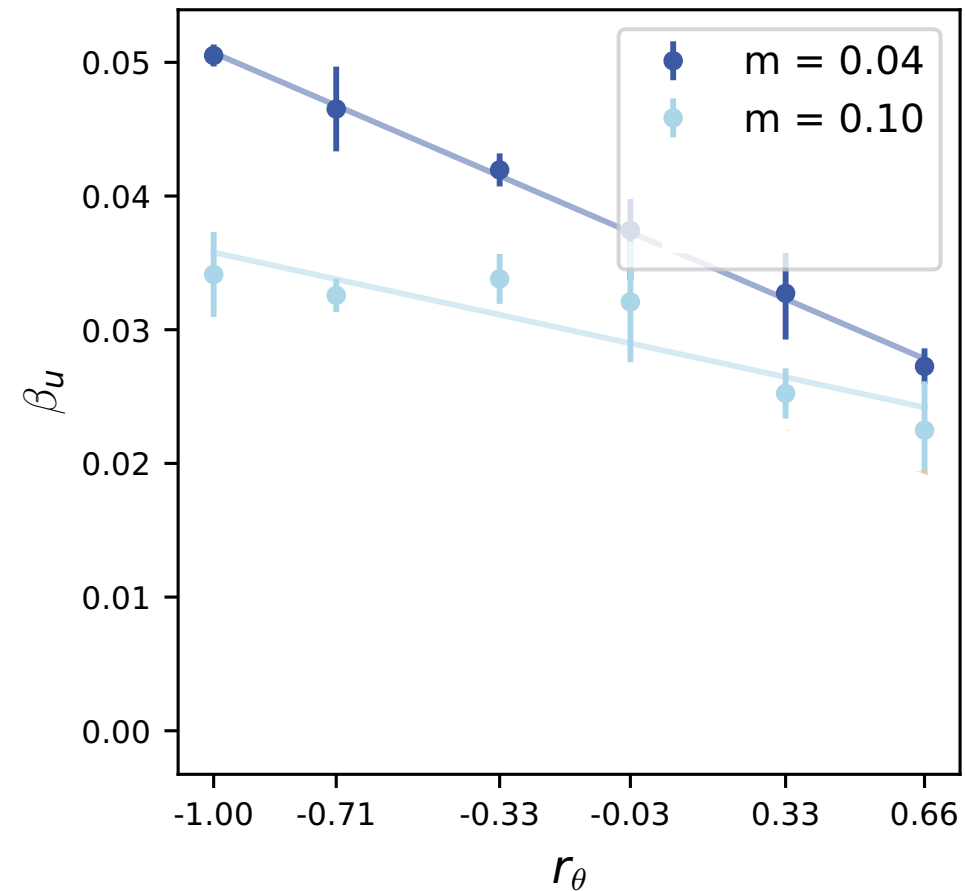
$$r_\theta \approx 0$$



$$r_\theta \approx 1$$

Environmental assortativity $r_\theta \sim$ Environmental spatial autocorrelation

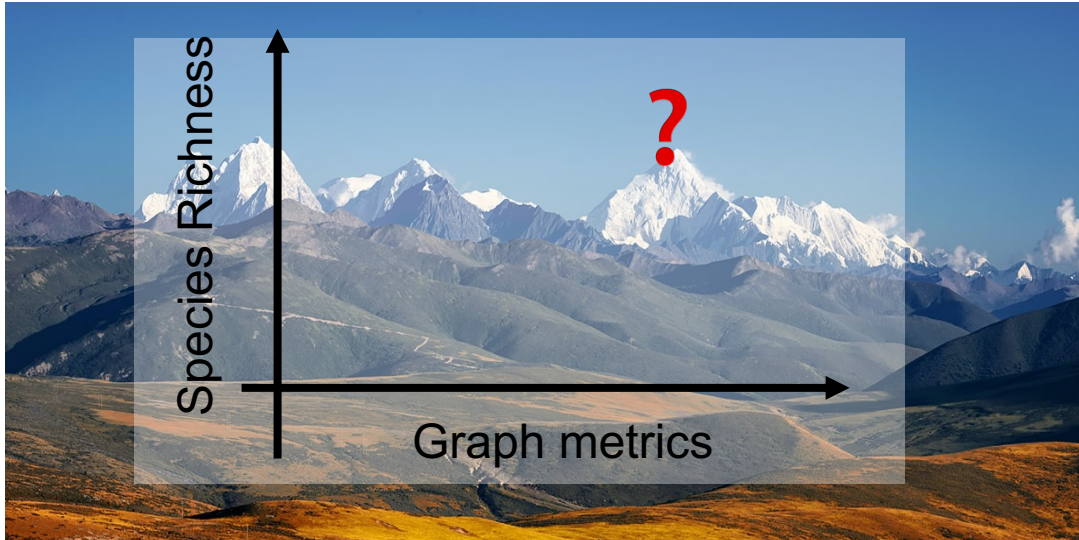
Experiment #2 – Effect of environmental assortativity on neutral differentiation



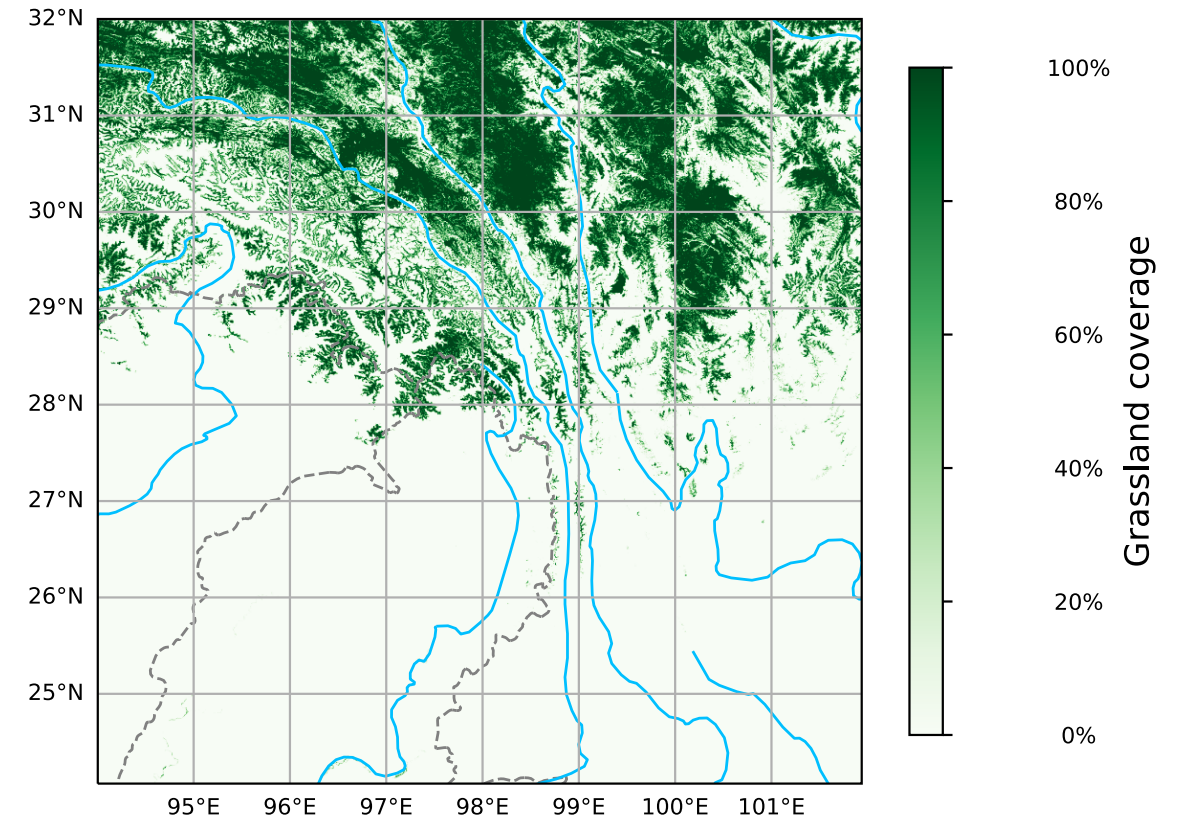
Boussange, V., & Pellissier, L. (2021). Topology and habitat assortativity drive neutral and adaptive diversification in spatial graphs. *BioRxiv*.

In revision @ **Communications Biology**

Theory validation and application



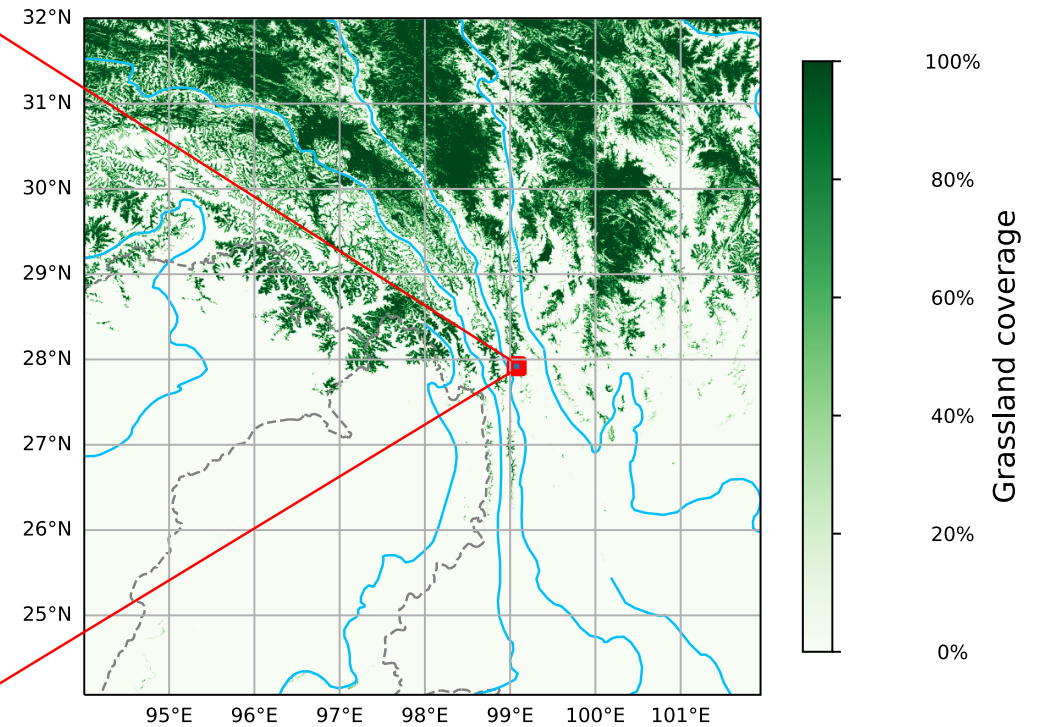
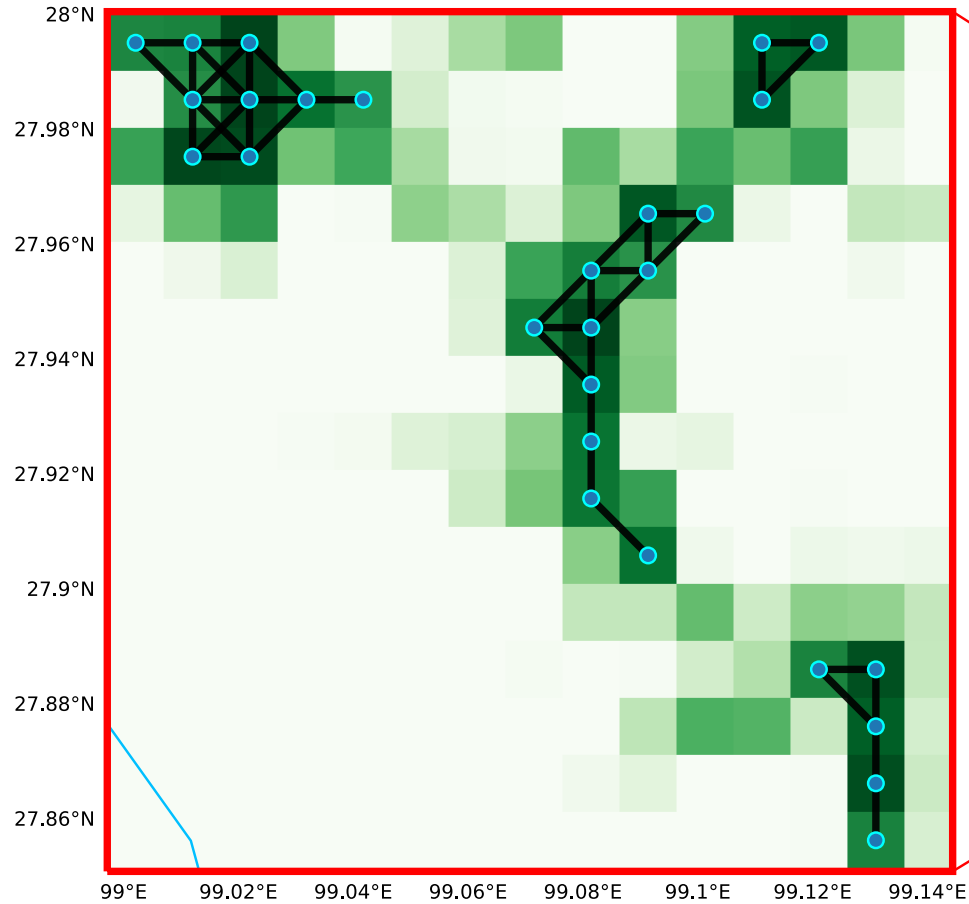
→ Project real landscapes on graphs



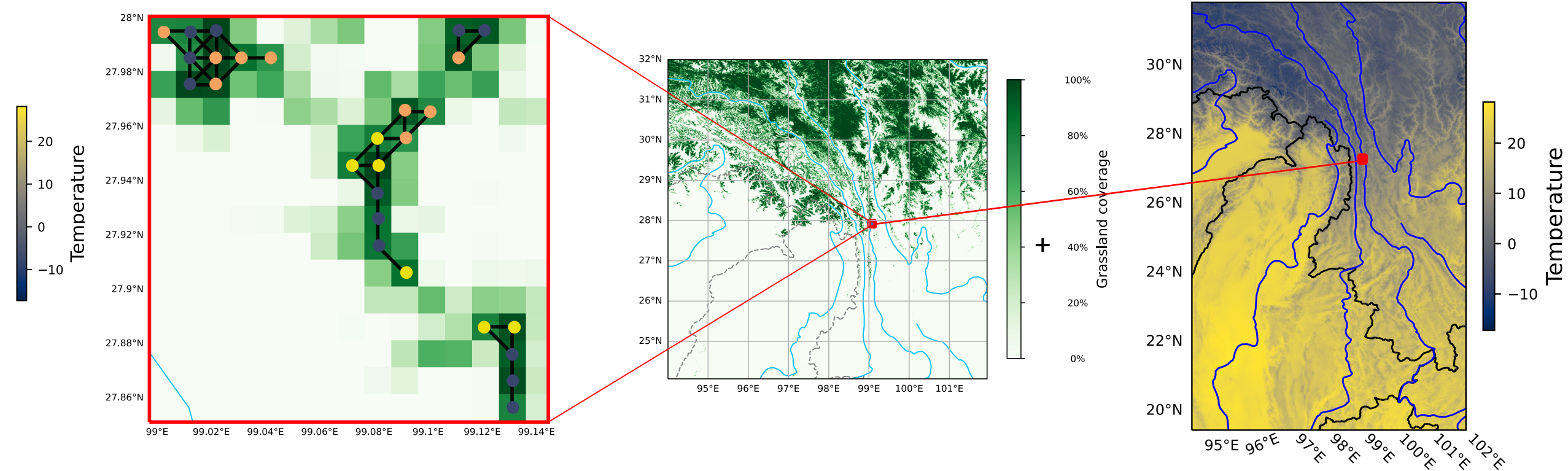
A global map of terrestrial habitat types, M. Jung et al. 2020

spatially explicit characterization of 47 terrestrial habitat types at 0.01° resolution, as defined in the (IUCN) habitat classification scheme

Graph representation

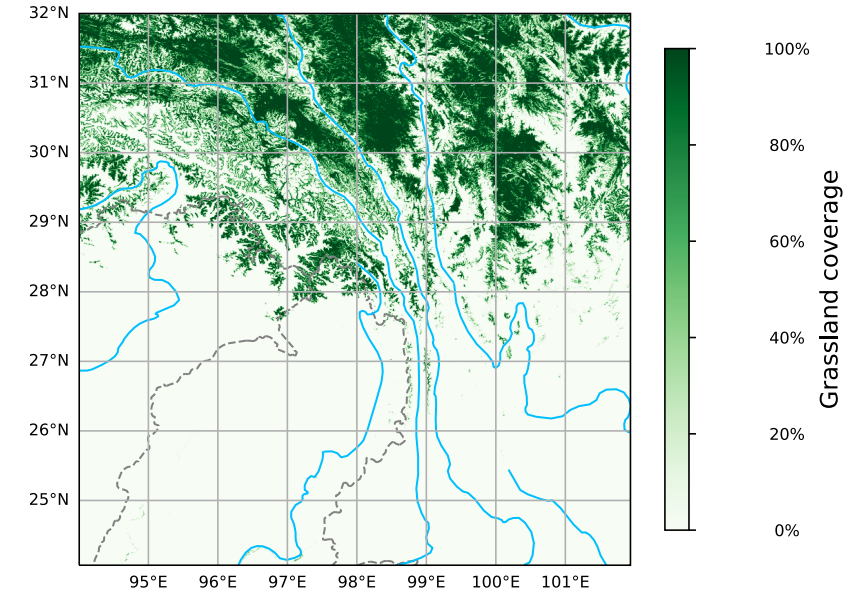


Graph representation with environmental conditions



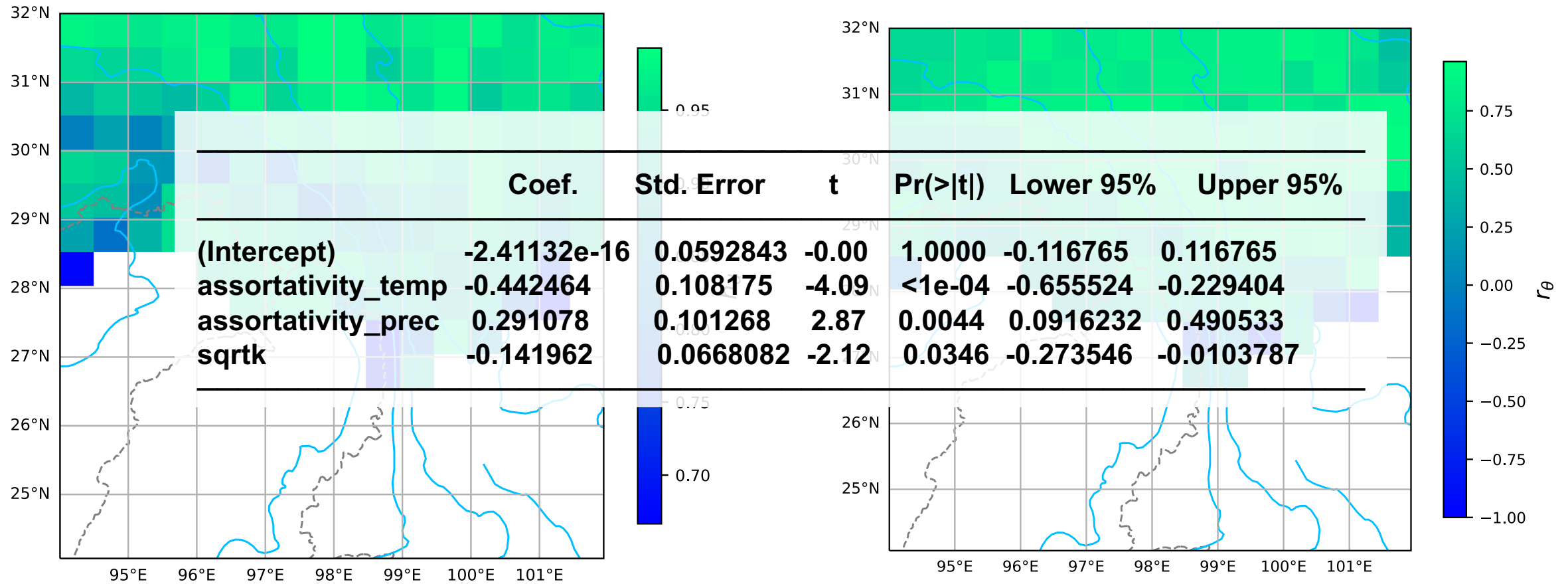
Systematic mapping of graph-based metrics

- Compute metrics for all graphs within window $0.25^\circ \times 0.25^\circ$
- Average
- Reproject on map with pixel size $0.25^\circ \times 0.25^\circ$
- Correlate with empirical data



Heterogeneity in degree

Temperature assortativity



Acknowledgements



Loïc Pellissier



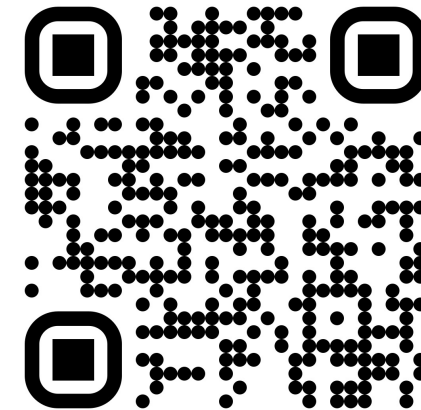
Yaquan Chang

LANDSCAPE
ECOLOGY



Thanks!
(looking for a position
next year 😊)

Check out my personal website



to discover more
about my research